Marketing A/B Testing Analysis

Executive Summary

Conversion Lift

+43%

Higher than Control Group (PSA)

Success Confirmed

Ad campaign highly successful and statistically significant (p-value < 0.05).

Strategic Action

Recommendation: Approve full campaign rollout based on validated results.

Data Integrity & Methodology

Data Overview

Total Users

588,101

Ad Group
564,577 (96%)
PSA Group (Control)
23,524 (4%)

Methodology

Chi-Square Test

Statistical approach used to compare the proportions of converted users between the control and test groups, ensuring results are not due to random chance.

Conclusion (Logistic Regression): Even after controlling for the time variable, being in the Ad group still significantly increases the conversion rate.

Key Performance Metrics

Control (PSA)

1.78%

Baseline

Lift

+43%

Test (Ad)

2.55%

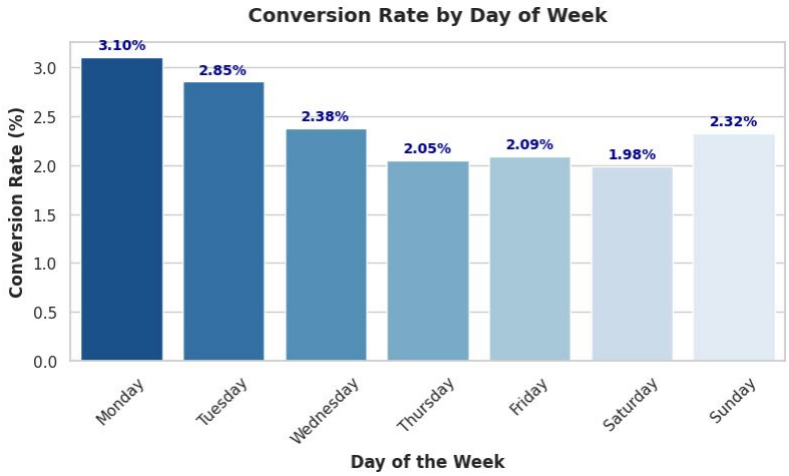
Optimized

Result is **Statistically Significant** (p-value < 0.05)

Temporal Performance Insights

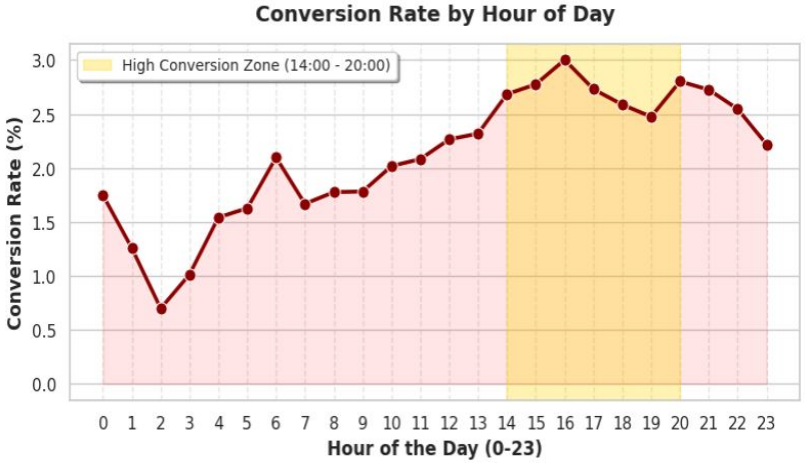
Conversion by Day of Week

Performance is **Highest on Monday & Tuesday**, with a gradual decline towards the weekend.



Conversion by Hour of Day

Peak performance occurs during the Afternoon/Evening hours (14:00 - 20:00).



Actionable Next Steps

Full Campaign Rollout

Approve immediate expansion based on the +43% conversion lift.

Budget Optimization

Shift more budget to Monday and Tuesday to capture peak performance periods.

Key Findings Recap

- Results are statistically significant with a p-value < 0.05.
- Peak performance occurs consistently during Afternoon/Evening hours.